

Java Package

A **java package** is a group of similar types of classes, interfaces and sub-packages.

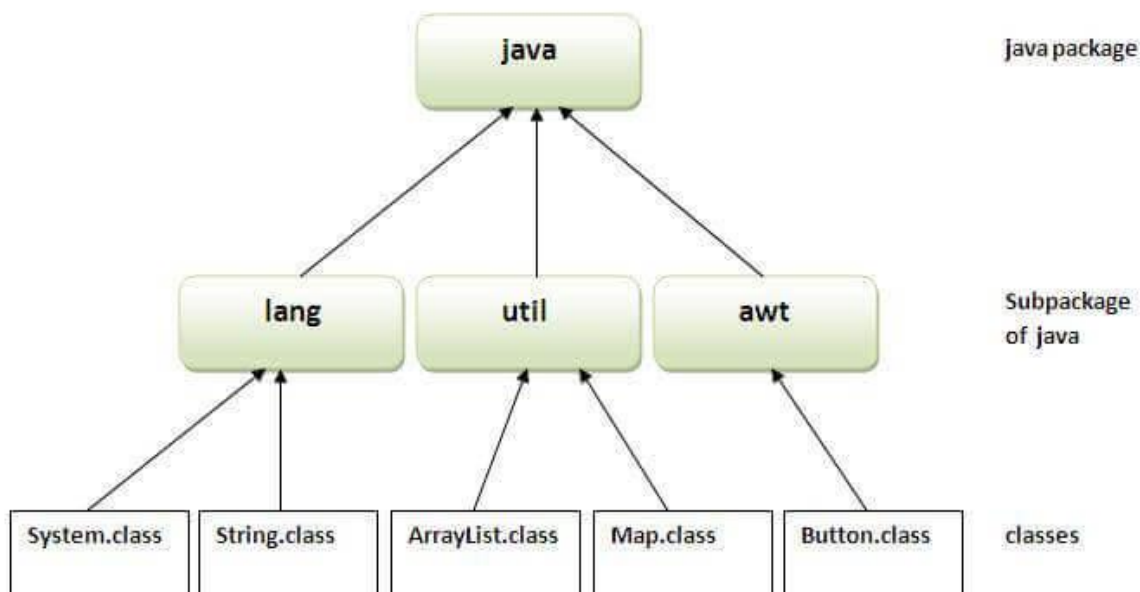
Package in java can be categorized in two form, built-in package and user-defined package.

There are many built-in packages such as java, lang, awt, javax, swing, net, io, util, sql etc.

Here, we will have the detailed learning of creating and using user-defined packages.

Advantage of Java Package

- 1) Java package is used to categorize the classes and interfaces so that they can be easily maintained.
- 2) Java package provides access protection.
- 3) Java package removes naming collision.



Simple example of java package

The **package keyword** is used to create a package in java.

```
package Series;
public class Square
{
    public void printSeries(int n)
    {
        System.out.print("Square Series: ");
        for (int i = 1; i <= n; i++)
        {
            System.out.print((i * i) + " ");
        }
        System.out.println();
    }
}
```

```
package Series;

public class Cube
{
    public void printSeries(int n)
    {
        System.out.print("Cube Series: ");
        for (int i = 1; i <= n; i++)
        {
            System.out.print((i * i * i) + " ");
        }
        System.out.println();
    }
}
```

```
package Series;
```

```
public class Fibonacci
{
    public void printSeries(int n)
    {
        int a = 0, b = 1;
        System.out.print("Fibonacci Series: " + a + " " + b);
        for (int i = 2; i < n; i++)
        {
            int next = a + b;
            System.out.print(" " + next);
            a = b;
            b = next;
        }
        System.out.println();
    }
}
```

How to access package from another package?

There are three ways to access the package from outside the package.

1. `import package.*;`
2. `import package.classname;`
3. fully qualified name.

1) Using packagename.*

If you use `package.*` then all the classes and interfaces of this package will be accessible but not subpackages.

The `import` keyword is used to make the classes and interface of another package accessible to the current package.

2) Using packagename.classname

If you import `package.classname` then only declared class of this package will be accessible.

Example of package by import package.classname

```
import Series.Fibonacci;
import Series.Cube;
import Series.Square;
import java.util.Scanner;

public class MainProgram
{
    public static void main(String[] args)
    {
        Scanner scanner = new Scanner(System.in);

        System.out.print("Enter the number of terms: ");
        int n = scanner.nextInt();

        // Instantiate objects of each class
        Fibonacci fibonacci = new Fibonacci();
        Cube cube = new Cube();
        Square square = new Square();

        // Print the series
        fibonacci.printSeries(n);
        cube.printSeries(n);
        square.printSeries(n);

        scanner.close();
    }
}
```

3) Using fully qualified name

If you use fully qualified name then only declared class of this package will be accessible. Now there is no need to import. But you need to use fully qualified name every time when you are accessing the class or interface.

It is generally used when two packages have same class name e.g. java.util and java.sql packages contain Date class.

Example of package by import fully qualified name

```
public static void main(String args[])
{
    pack.A obj = new pack.A();//using fully qualified name
    obj.msg();
}
```